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Dialysis

Mass Transfer Discussion

In this paper, mathematical models of hemodialysis are discussed to understand how system efficiency can be improved and predicted analytically. A countercurrent dialysis system includes blood flow and dialysate flow. During the flow, toxin particles are transported from the blood region through a membrane to the dialysate region. Thus, the study of mass-transfer in a dialysis system is crucial as it helps remove the toxins from the patient's blood to the dialysate flow which can be discarded easily afterward.

$$\rho_{i,j} \nabla \vec{u}_{i,j} = - s_{i,j} \tag{1}$$

Where:

ρ: density

∇: Hamilton operator

$\vec{u}$: $velocity$

s: source or sink

The first equation demonstrates the conservation of mass principle. The subscript *i* denotes blood and dialysate and *j* denotes flow regions. The left side of the equation has a density times a velocity gradient, which is taken from the continuity equation. When moving the density symbol into side the Hamilton operator, a gradient of mass flux is presented $\nabla\left(\vec{u}_{i,j} p_{i,j}\right)$. A mass flux represents the amount of mass being transported through an area per unit time. This means that the left-hand side of the equation demonstrates the mass flux of the blood and dialysate sides. In a hemodialysis system, toxin mass is being transferred from the blood flow to the dialysate flow. Thus, the equation must also consider the added or subtracted amount of mass into the flow. Especially, the symbol *s* becomes a sink on the blood side but a source on the dialysate side.

$$`\nabla\left(\vec{u}_{i,j} c_{i,j}\right) = \nabla\left(D_{i,j} \nabla c_{i,j}\right) - g_{i,j} \quad (2)$$

Where:

c: toxin concentration

D: diffusion coefficient of toxins

G: solute source/sink

Equation 2 is very similar to equation 1, the biggest difference is equation 1 demonstrates the mass flux in blood and dialysate flows while equation 2 focuses on the toxin concentration. The left-hand side of equation 2 represents a concentration flux. As shown, the density ρ symbol is replaced with a concentration c symbol. This is again derived from the conservation of mass principle. A new term is introduced, the diffusion coefficient of toxins D demonstrates the quantity of toxin being diffused from the blood region to another passes through a unit area per unit time assuming the volume-concentration gradient is unity. The g symbol is the toxin source or sink depending on which side is being analyzed. Note that the g symbol contains all solute transport caused by both diffusion and convection across the membrane.

The main purpose of hemodialysis is to clear out the toxin particles flowing in the blood vessels. Thus, the performance and efficiency of an artificial kidney are measured by the clearance level of toxins. An analytical way to figure out the amount of toxins being transported is by using the equation below.

$$c_L = \frac{\left(Q_{bi} c_{bi} - Q_{bo} c_{bo}\right)}{c_{bi}} \quad (3)$$

Where:

C_L: clearance of toxins

Q_{bi}: blood flow rates at the inlet

Q_{bo}: blood flow rates at the outlet

c_{bi}: toxin concentrations at the blood inlet

c_{bo}: toxin concentrations at the blood outlet

It is reasonable to include only the factors related to the toxin concentration and blood flow rates as the main goal is to determine the toxin clearance level. When multiplying a volume flow rate with a molar concentration "Qc", a molar concentration flow rate is created. Thus, the

$Q_{bi}c_{bi}$ and $Q_{bo}c_{bo}$ terms indicate the toxin concentration flow rate at the blood inlet and outlet, respectively. When subtracting the concentration flow rate the inlet by the concentration flow rate at the out, the result provides the change in concentration throughout the blood region. Then, this change of concentration flow rate is divided by the initial concentration to compare how significant the change is compared to the initial concentration. The value of the c_L term shall be a ratio with a unit of $\frac{1}{t}$ where t is time (often used as in second). This c_L value shows how much of the toxin is being transported out of the blood flow per unit time. Thus, the clearance of toxin indicates how efficient the performance of the hemodialysis is.

Efficiency Improvement

Based on the simulation and analysis of the article, the inlet and out geometrical structures of the dialysate side affect the unity of the flow fields. This holds for the geometrical structures of the blood inlet and outlet but at a less significant level. The more uniform the flow fields are, the more toxins are transported. Therefore, the first suggestion to improve the hemodialysis efficiency is to select the correct geometrical structures of the dialysate inlet and outlet. Different geometrical structures should be tested and analyzed if there is not an experiment or simulation performed yet.

The article also notes that the blood and dialysate flow rates affect mass transport even more significant compared to the geometrical structures of the flow inlets and outlets. The real flow profiles showed a smaller creatinine clearance when the blood flow rate was high or the dialysate flow rate was low. Thus, another suggestion is to ensure either the blood flow rate is low or the dialysate is high. The third idea for efficiency improvement is to implement ultrafiltration in the countercurrent dialysis system. Ultrafiltration applications can quickly improve the mass transfer efficiency by transporting the solute in diffusion and convection simultaneously. This proposal is analyzed in the second article and the section below.

Mathematical Model

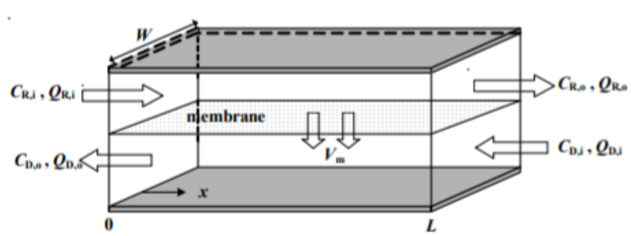

Figure 1: Parallel-plate Countercurrent Dialysis System

The last suggestion for the hemodialysis efficiency improvement is discussed in this second article. It is noted that the ultrafiltration reduces the treatment time and is tolerated by patients. Figure 1 is taken from the article to demonstrate the subscripts of the variables more easily. The governed equation utilized in this article is

$$-\frac{dQ_R}{Wdx} = -\frac{dQ_D}{Wdx} = \vec{V}_m \quad (4)$$

Where:

Q_R: volume flow rate of the upper channel

Q_D: volume flow rate of the bottom channel

W: width of the flat-plate

$\vec{V}_m$: average ultrafiltration flux

The subscripts *i* and *o* indicate the inlet and outlet respectively. As shown in figure 1, as the upper flow moves from left to right, the bottom flow moves in the opposite direction. This countercurrent flow system has shown higher energy and mass transfer efficiency. Across the membrane placed between the upper and lower regions, the ultrafiltration flux as being transported as the pressure decreases along the conduit length. Average transmembrane ultrafiltration and pressure are used to simplify the equation. Based on the general mass balance equations, the equation for the solute in both flows is derived in the form of the equation above. This equation emphasizes that the rate of change of volume flow rate is equal to the flux of our constituent across the membrane.

There are two elements considered when determining the efficiency improvement: overall mass transfer rate *M*, separation efficiency η and percent difference *E%.* The improvement was incorporated mathematically by the following equations.

$$M = Q_{R,i}C_{R,i} - Q_{R,o}C_{R,o} = Q_{D,i}C_{D,i} - Q_{D,o}C_{D,o} \quad (5)$$

The subscripts *R* and *D* present the upper flow and bottom flow respectively. Equation 5 simply states that the overall mass transfer rate in a dialyzer must equal to the change in the concentration flow rate of the solute either in the upper or lower region. This is again derived from the principle of mass conservation.

$$= \frac{M}{LWK(C_{R,i} - C_{D,i})} \quad (6)$$

The $LWK(C_{R,i} - C_{D,i})$ term is the mass transfer rate when the dialysate and retentate concentrations remain the same as at the inlet concentration without ultrafiltration. The separation efficiency η is defined as the actual mass transfer rate to the maximum possible mass transfer rate by only diffusion.

$$E\% = \frac{- \quad {}_{=0}}{{}_{=0}} \times 100\% \qquad (7)$$

Finally, the efficiency improvement of the ultrafiltration application is found using equation 7. The percentage is calculated by finding the difference in separation efficiency of the dialysis system, one is coupled with an ultrafiltration while one is not. This difference is then divided by the separation efficiency of the system without the ultrafiltration to demonstrate how the significant percentage increase in mass transfer results from the ultrafiltration application. Please note that all the information, figure, data and equations were obtained from the two papers listed in the reference section.

Reference

Ding, W., Li, W., Sun, S., Zhou, X., Hardy, P. A., Ahmad, S., & Gao, D. (2015). Three-Dimensional Simulation of Mass Transfer in Artificial Kidneys. *Artificial Organs*, *6*, 79. https://doi-org.lopes.idm.oclc.org/10.1111/aor.12415

Tu, J.-W., Ho, C.-D., & Chuang, C.-J. (2009). Effect of ultrafiltration on the mass-transfer efficiency improvement in a parallel-plate countercurrent dialysis system. *Desalination*, *242*(1), 70–83. https://doi-org.lopes.idm.oclc.org/10.1016/j.desal.2008.03.032